Predicting Disease of Tomato Leaves

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Problem: Tomato Plant Diseases in Agriculture

Potatoes and tomatoes are the most commonly consumed vegetables (Kantor). Yet, more than 25% of total tomato crop production is lost annually because of plant diseases that have significantly reduced food production and degrade crop quality.

However, human identification of plant diseases can be slow and inefficient. Therefore, the use of machine learning is better as it can quickly identify diseases in agriculture.

Background: Machine Learning Methods for Disease Detection

Previous Methods of Detection:

- Support Vector Machines (SVM) have a 60-70% accuracy
  - Lowest accuracy
- ANN's have a 80-85% accuracy

2011–2012:

- OTSU method, Gabor filters & ANN have a 91% accuracy

2018:

- CNN using VGG have a 99.58% accuracy
  - Highest accuracy

CNN's repeatedly outperformed previous methods of detection.

Data:

- We found our dataset, titled "**Tomato Leaf Disease Detection,**" on Kaggle. It contains two sets of images, a training set and a validation set.

- The image sets contain images of tomato leaves with nine different diseases and healthy leaves as well.

Tomato leaf with **bacterial spots**

Tomato leaf with **Yellow Leaf Curl Virus**

Healthy tomato leaf

Analysis

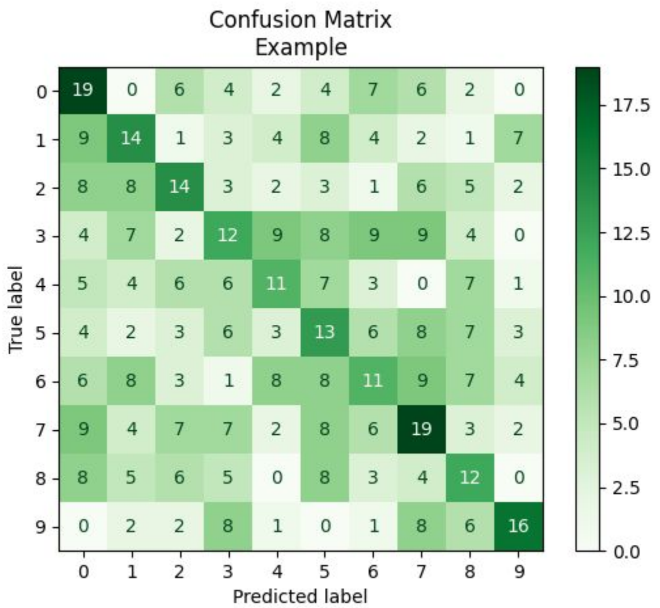
The data is described through the different types of diseases for tomato leaves:

- Tomato Mosaic Virus
- Target Spot
- Bacterial Spot
- Tomato Yellow Leaf Curl Virus
- Late Blight
- Leaf Mold
- Early Blight
- Spider mites Two-Spotted Spider Mites
- Healthy
- Septoria Leaf Spot

CNN Model Training & Solution

- We used the tomato images from kaggle to train the CNN to recognize different types of tomato leaf diseases.
- We split the existing dataset into 80% training, 10% validation, and 10% testing
- We predict the possible disease of the tomato leaf based on an image of the leaf

Evaluation



Per-Class Metrics:

- Precision: Measures how many images classified positively are actually part of the class
- Recall: Measures how many images that were part of the class that got classified correctly
- Matthews Correlation Coefficient: Measures how strongly correlated the model's classifications are with the actual classifications

Overall Metrics:

- Accuracy: Total % of correctly classified images

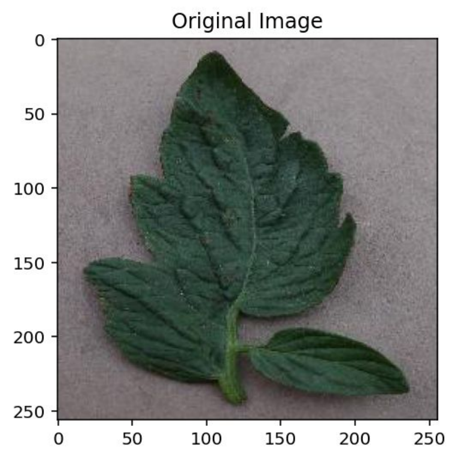
Other Evaluation Tools:

- Confusion Matrix: Displays the differences between the predicted classifications and the actual classifications

Our CNN Model

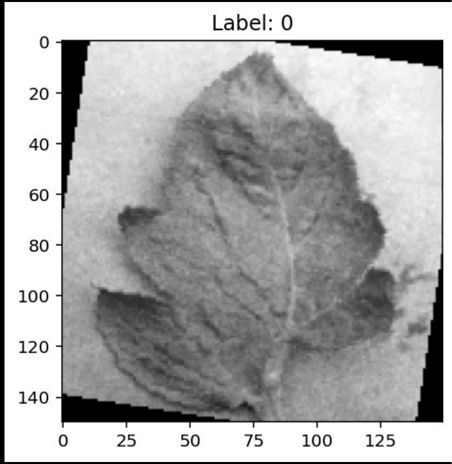
Preprocessing

- **Grayscale**
  - reduces complexity and noise
- **Resize:**
  - input size is consistent(150×150)
- **Rotation:**
  - improves generalization and prevents overfitting
  - Black are caused by rotation (not an error)
- **Normalization**
  - scales pixels values



Original Image (RGB, variable size)

Preprocessing pipeline



Processed Image (Grayscale, Resized, Augmented)

CNN Architecture

- 8 Convolutional Layers (Deep)
- Batch Normalization
  - Stabilizes gradients, reduces overfitting
- ReLU Activation
  - Introduce non-linearity
- MaxPooling
  - Reduces spatial dimensions by half
- Flattening
  - 3D vector → 1D vector for classifier

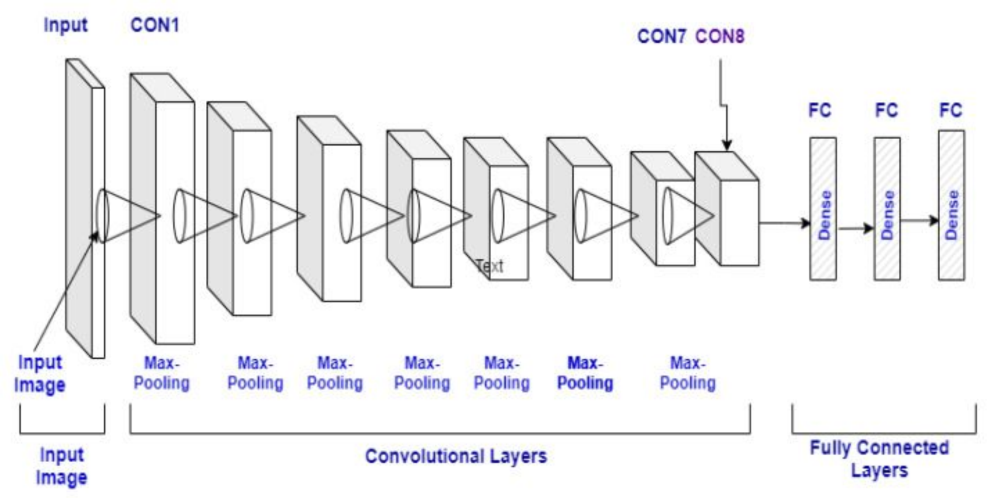


Figure 1: CNN Model Architecture

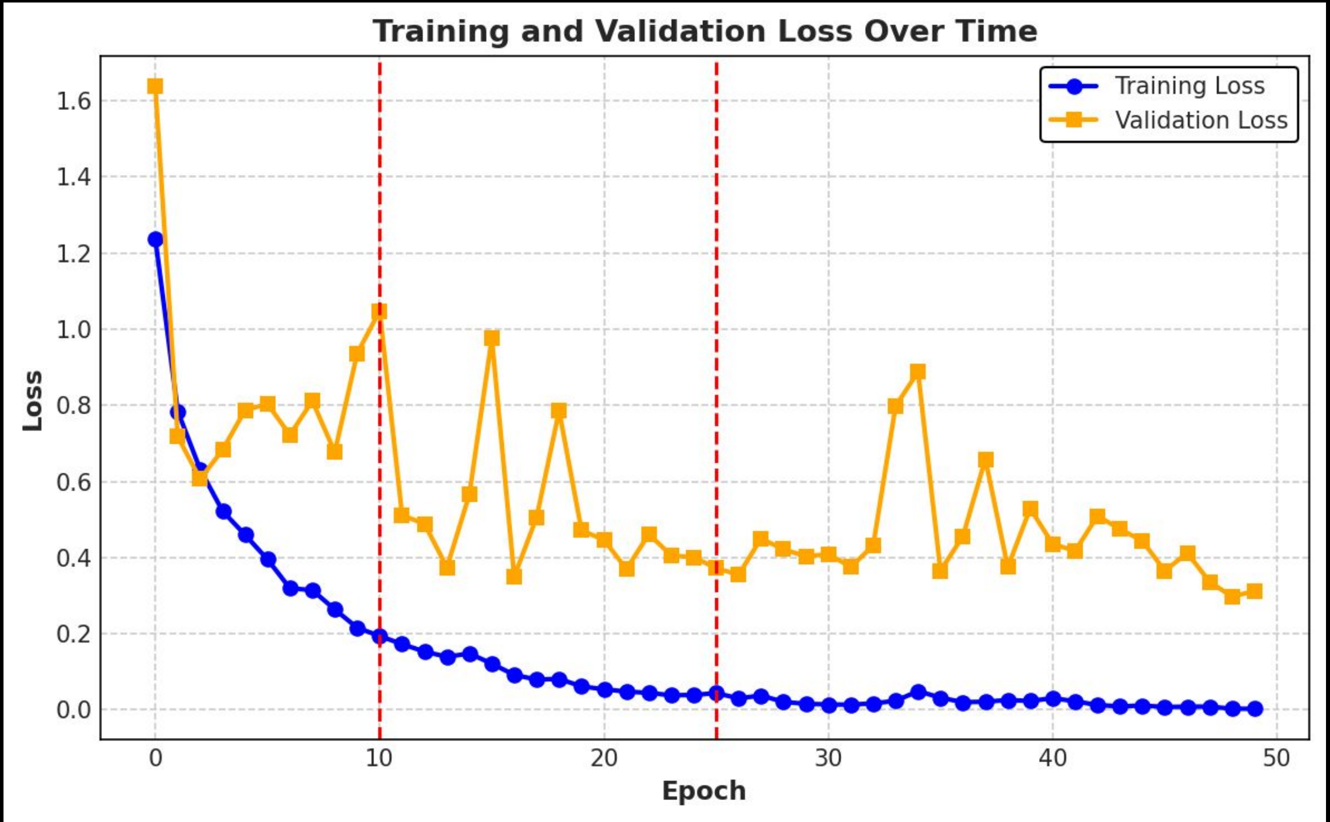
Training Loop

- GPU training when available, CPU otherwise
- Standard Pytorch training loop
  - Reset gradients
  - Forward pass
  - Compute loss
  - Compute gradients
  - Update weights
- Tracks training and validation losses at each epoch
  - Used for plotting

Training Parameters

- Model: nn.Module
- Epochs (10, 25, 50)
- Loss Function
- Optimizer
  - SGD with momentum, Adam, AdamW
- Data Loaders
  - For training and validation sets
- Dropout
  - 0.5

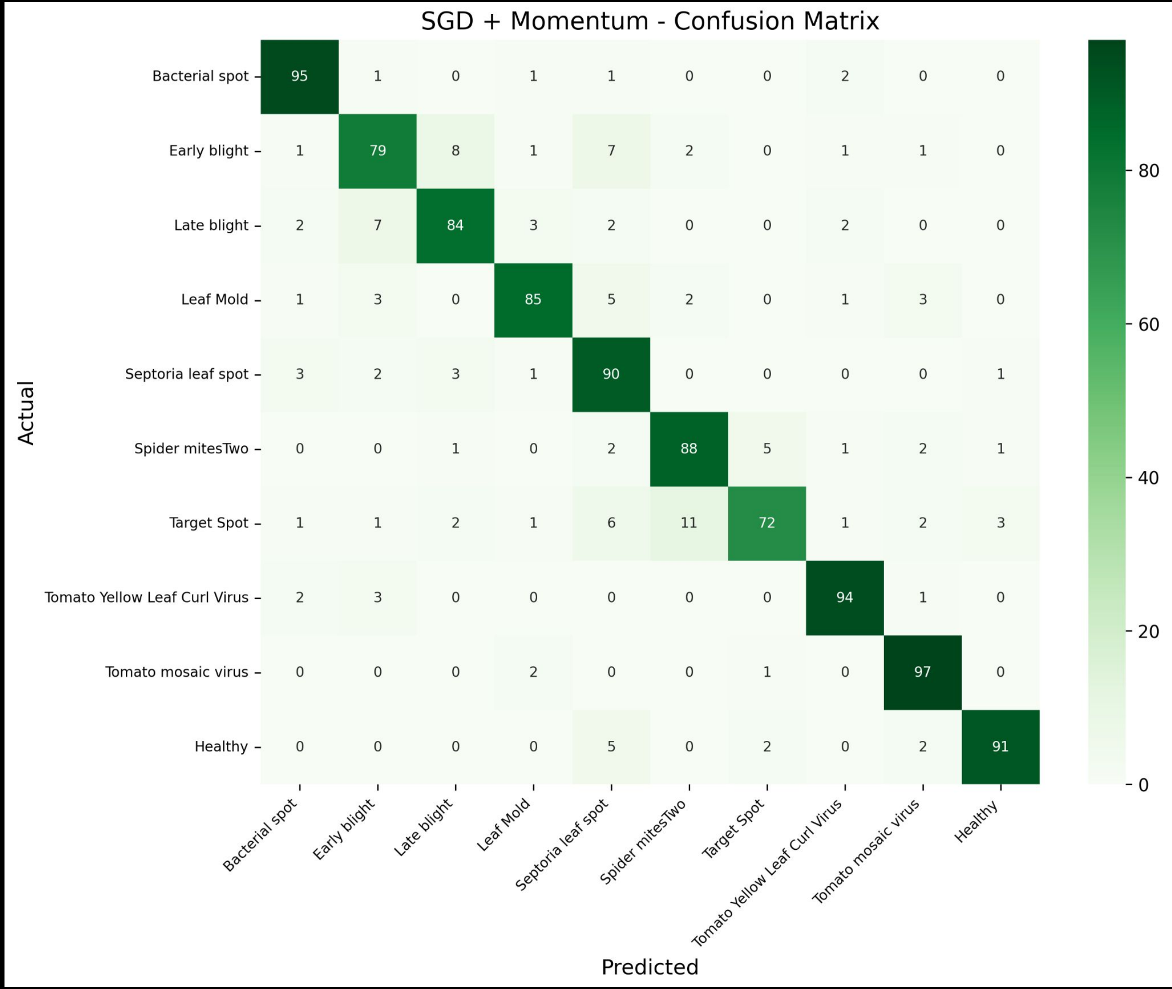
Results: SGD with Momentum



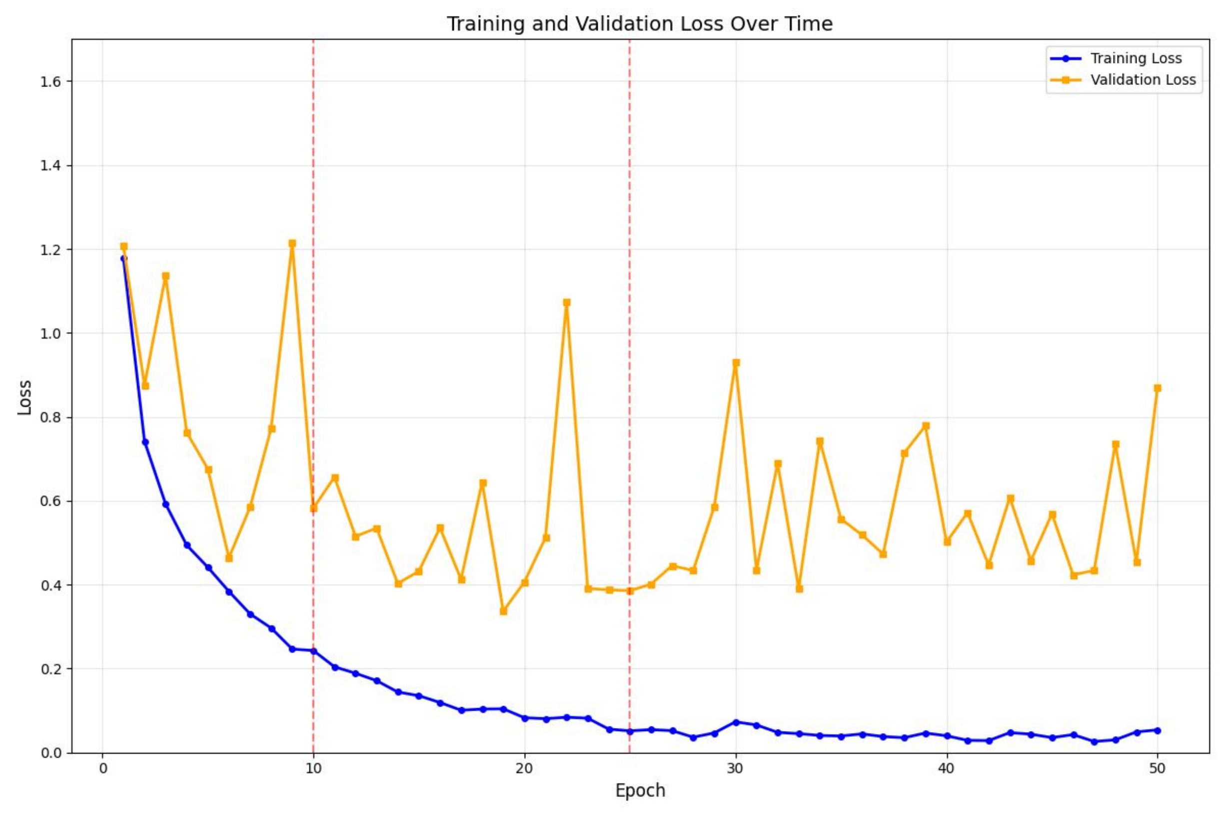
SGD

Epochs	Accuracy
10	0.839
25	0.849
50	0.872

Class	Precision	Recall	MCC
Bacterial spot	0.898	0.970	0.926
Early blight	0.806	0.790	0.776
Late blight	0.865	0.830	0.831
Leaf Mold	0.884	0.840	0.847
Septoria leaf spot	0.765	0.880	0.799
Spider mites Two-spotted spider mite	0.871	0.880	0.862
Target Spot	0.901	0.730	0.793
Tomato Yellow Leaf Curl Virus	0.913	0.940	0.918
Tomato Mosaic Virus	0.896	0.950	0.914
Healthy	0.938	0.910	0.916


SGD + Momentum - Confusion Matrix
Actual
Predicted
Bacterial spot
Early blight
Late blight
Leaf Mold
Septoria leaf spot
Spider mitesTwo
Target Spot
Tomato Yellow Leaf Curl Virus
Tomato mosaic virus
Healthy
95 1 0 1 1 0 0 2 0 0
1 79 8 1 7 2 0 1 1 0
2 7 84 3 2 0 0 2 0 0
1 3 0 85 5 2 0 1 3 0
3 2 3 1 90 0 0 0 0 1
0 0 1 0 2 88 5 1 2 1
1 1 2 1 6 11 72 1 2 3
2 3 0 0 0 0 0 94 1 0
0 0 0 2 0 0 1 0 97 0
0 0 0 0 5 0 2 0 2 91
80
60
40
20
0

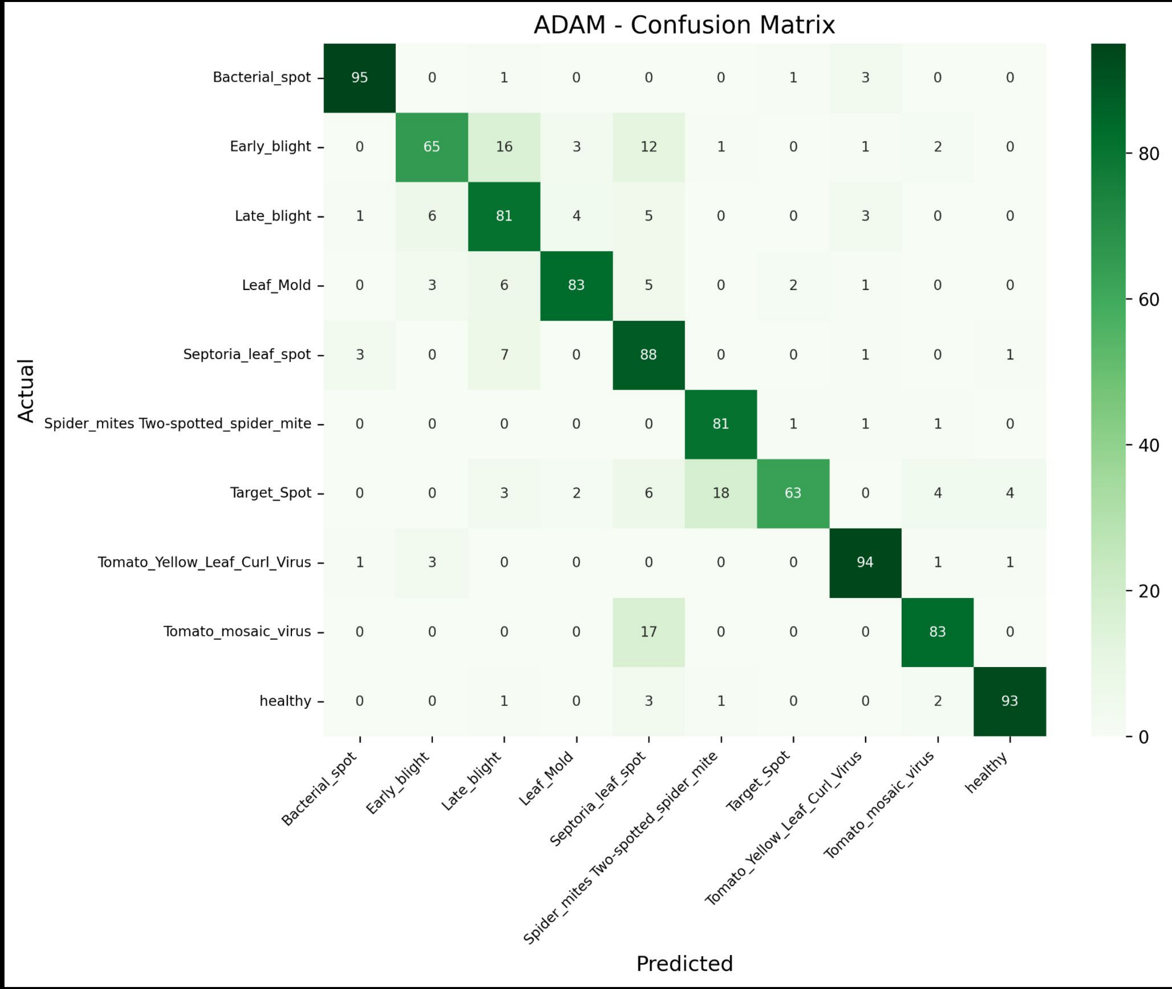
Results: Adam



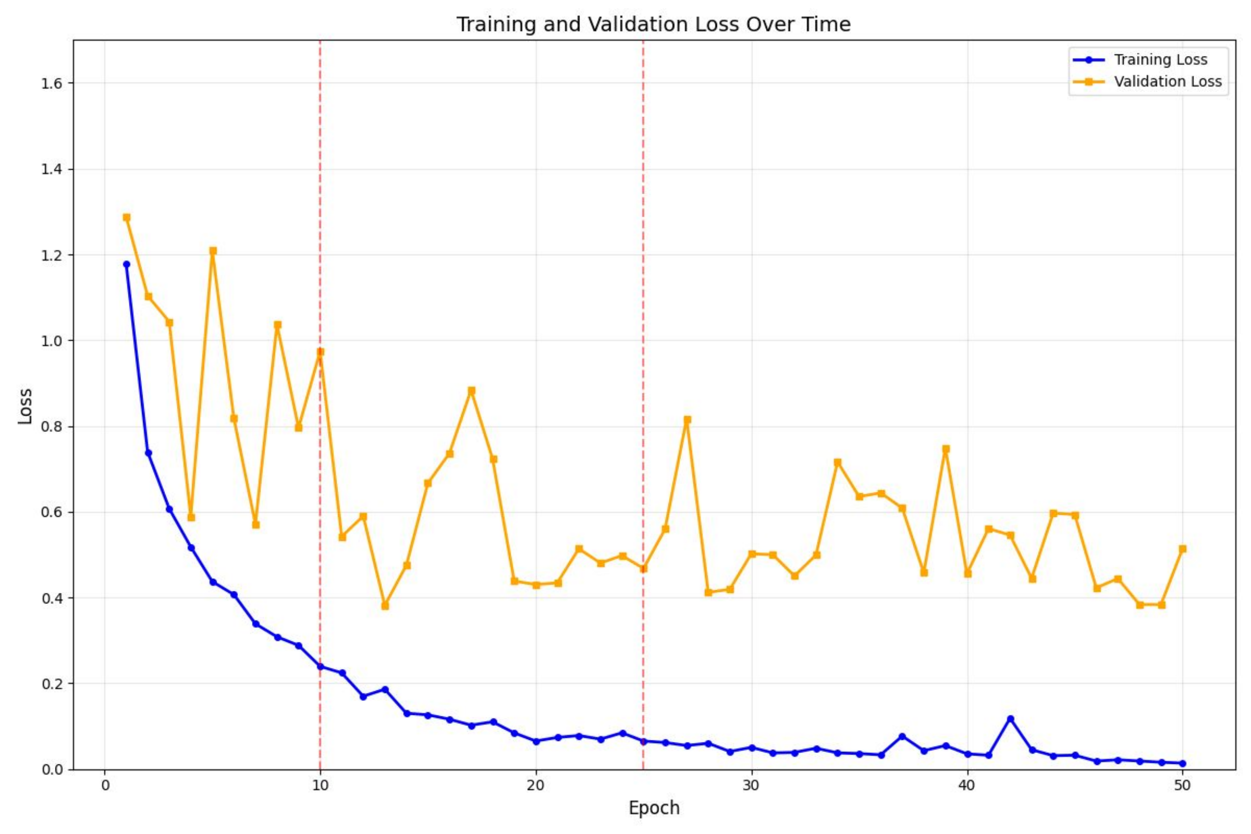
Adam

Epochs	Accuracy
10	0.799
25	<u>0.839</u>
50	0.798

Class	Precision	Recall	MCC
Bacterial spot	0.900	0.810	0.838
Early blight	0.549	0.890	0.658
Late blight	0.654	0.830	0.703
Leaf Mold	0.885	0.690	0.760
Septoria leaf spot	0.767	0.660	0.682
Spider mites Two-spotted spider mite	0.878	0.857	0.855
Target spot	0.859	0.670	0.735
Tomato Yellow Leaf Curl Virus	0.853	<u>0.930</u>	0.878
Tomato Mosaic virus	0.952	0.790	0.854
Healthy	<u>0.978</u>	0.870	<u>0.914</u>


ADAM - Confusion Matrix
Actual
Predicted
Bacterial_spot
Early_blight
Late_blight
Leaf_Mold
Septoria_leaf_spot
Spider_mites Two-spotted_spider_mite
Target_Spot
Tomato_Yellow_Leaf_Curl_Virus
Tomato_mosaic_virus
healthy
95 0 1 0 0 0 1 3 0 0
0 65 16 3 12 1 0 1 2 0
1 6 81 4 5 0 0 3 0 0
0 3 6 83 5 0 2 1 0 0
3 0 7 0 88 0 0 1 0 1
0 0 0 0 0 81 1 1 1 0
0 0 3 2 6 18 63 0 4 4
1 3 0 0 0 0 0 94 1 1
0 0 0 0 17 0 0 0 83 0
0 0 1 0 3 1 0 0 2 93
80
60
40
20
0

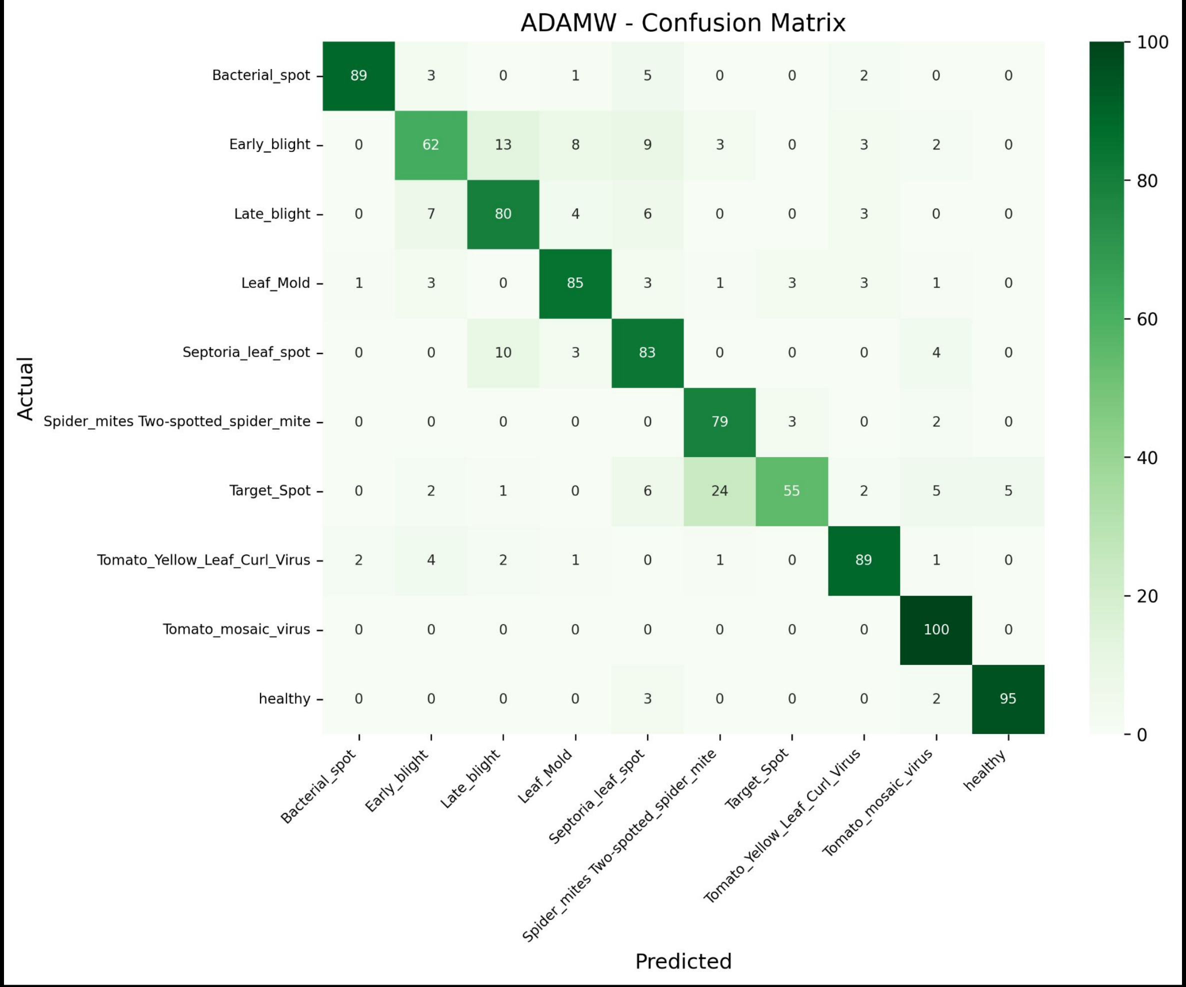
Results: AdamW



AdamW

Epochs	Accuracy
10	<u>0.830</u>
25	0.796
50	0.796

Class	Precision	Recall	MCC
Bacterial spot	<u>0.967</u>	0.890	0.920
Early blight	0.765	0.620	0.658
Late blight	0.755	0.800	0.751
Leaf Mold	0.833	0.850	0.824
Septoria leaf spot	0.722	0.830	0.747
Spider mites Two-spotted spider mite	0.731	0.940	0.812
Target spot	0.902	0.550	0.681
Tomato yellow leaf curl virus	0.873	0.890	0.868
Tomato mosaic virus	0.855	<u>1.000</u>	0.916
Healthy	0.950	0.950	<u>0.944</u>


ADAMW - Confusion Matrix
Actual
Predicted
Bacterial_spot 89 3 0 1 5 0 0 2 0 0
Early_blight 0 62 13 8 9 3 0 3 2 0
Late_blight 0 7 80 4 6 0 0 3 0 0
Leaf_Mold 1 3 0 85 3 1 3 3 1 0
Septoria_leaf_spot 0 0 10 3 83 0 0 0 4 0
Spider_mites Two-spotted_spider_mite 0 0 0 0 0 79 3 0 2 0
Target_Spot 0 2 1 0 6 24 55 2 5 5
Tomato_Yellow_Leaf_Curl_Virus 2 4 2 1 0 1 0 89 1 0
Tomato_mosaic_virus 0 0 0 0 0 0 0 0 100 0
healthy 0 0 0 0 3 0 0 0 2 95
Bacterial_spot
Early_blight
Late_blight
Leaf_Mold
Septoria_leaf_spot
Spider_mites Two-spotted_spider_mite
Target_Spot
Tomato_Yellow_Leaf_Curl_Virus
Tomato_mosaic_virus
healthy
100
80
60
40
20
0

Final Results

- Best model used **SGD with momentum** and ran for 50 epochs.
- Test Accuracy: **87.2%**
- Best Class Precision: **Healthy** (0.938)
- Best Class Recall: **Bacterial Spot** (0.970)
- Best Class MCC: **Bacterial Spot** (0.926)

- Surprising, as we were expecting AdamW to have the best testing accuracy. If there were other parameter tunings tested our final results would have likely differed.
- One thing to note: For AdamW optimizer at 10 epochs, Tomato mosaic virus had perfect recall!

References

Source for the Dataset

Name: Tomato leaf disease detection using CNN
Owner: Kaustubh b
CC0: Public Domain
Link: https://www.kaggle.com/datasets/kaustubhb999/tomatoleaf

Source for Problem and Background Slides

Trivedi, N. K., Gautam, V., Anand, A., Aljahdali, H. M., Villar, S. G., Anand, D., Goyal, N., & Kadry, S. (2021). Early Detection and Classification of Tomato Leaf Disease Using High-Performance Deep Neural Network. *Sensors (Basel, Switzerland), 21*(23), 7987. https://doi.org/10.3390/s21237987

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Figure 1: CNN architecture model. Trivedi, N. K., Gautam, V., Anand, A., Aljahdali, H. M., Villar, S. G., Anand, D., Goyal, N., & Kadry, S. (2021, November 30). *Early detection and classification of tomato leaf disease using high-performance deep neural network*. Sensors (Basel, Switzerland). https://pmc.ncbi.nlm.nih.gov/articles/PMC8659659/figure/sensors-21-07987-f003/.

Thank you!